Weight of Expectation: Gendered Impacts of Body Image on Adolescent Well-being

Research Questions

Proportion: Does the proportion of students feeling "Sad or Hopeless" significantly differ from the 0.30 benchmark?

Mean: Does the average BMI Percentile (BMIPCT) significantly differ from the 65.0 national benchmark?

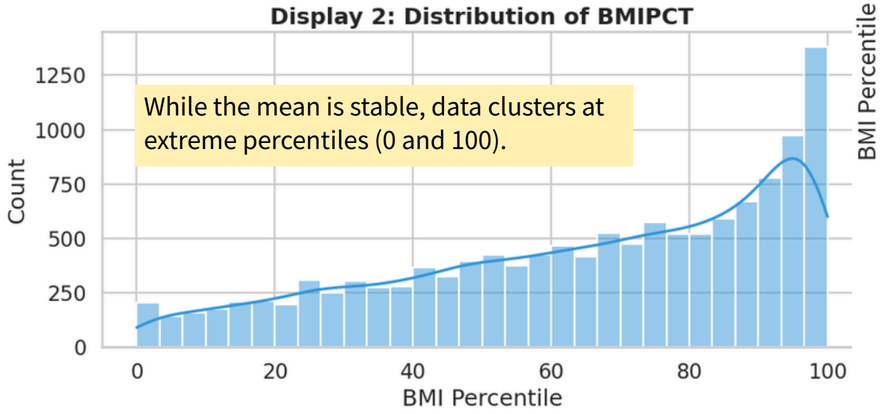
Data and Variables

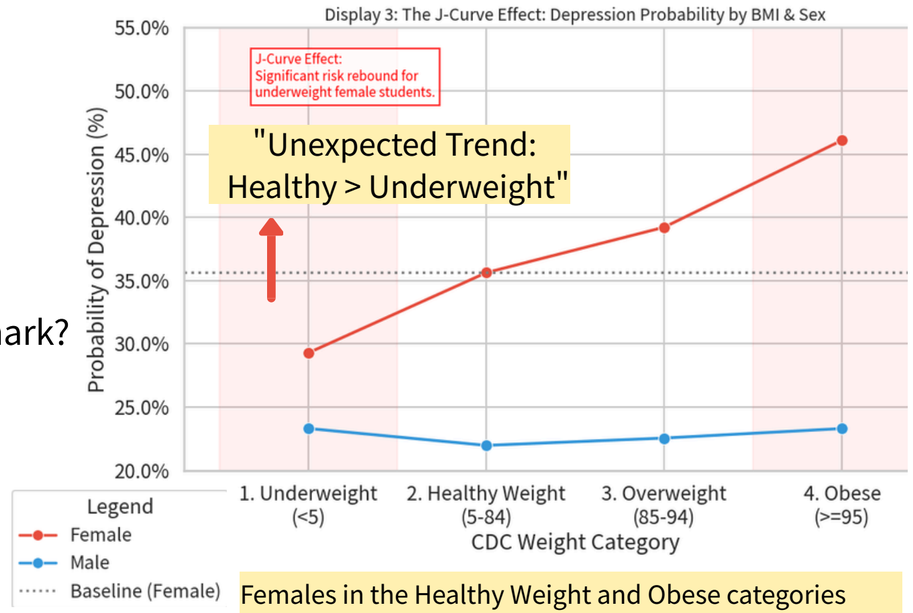
YRBS 2007 Dataset (n = 12,902)

SadOrHopeless-Binary(1 = Yes, 0 = No).

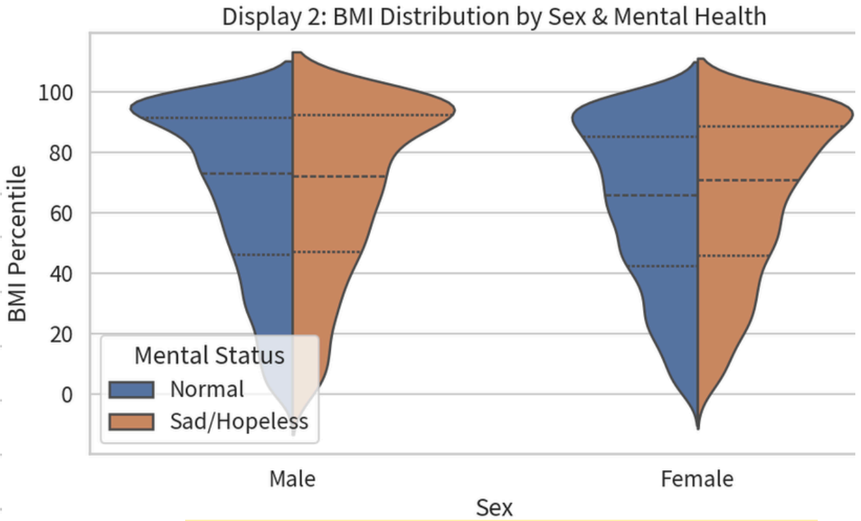
BMIPCT-Continuous

EDA Highlights





Females in the Healthy Weight and Obese categories report the highest depression probability, suggesting unique body image pressures even within normal ranges.



Females show a much stronger correlation between high BMI and psychological distress than males.

Methods & Results

Analysis	SadOrHopeless	BMIPCT
Method	Z-test	T-test
Benchmark	p_0=0.30	μ_0=65.0
Sample Stat	$\hat{p}$=0.2978	$\bar{x}$=64.82
CI	[0.2899, 0.3057] Contains 0.30	[64.35, 65.30] 65.0
P-value	**0.5891**	**0.4789**
	P > 0.05 Not Significant	
Decision	Fail to Reject H_0	Fail to Reject H_0

Conclusion

Statistical Verdict: At the 5% significance level, we failed to reject the null hypotheses for both variables. The sample does not significantly deviate from national benchmarks.

Core Insight:the J-Curve effect reveals hidden risks: extreme body types (underweight/obese) act as psychological thresholds, with females being more vulnerable to body image pressures